Owen's Extra Practice • week of Oct 24

Name: _____ Date: _____

This week: Complex fractions • Mixed numbers & dividing fractions • Unit rate & constant of proportionality (y = kx)

Show your work on scrap paper. Check: does the answer make sense?

1 Simplify the complex fraction:
(2 1/2) / (3/4)

Answer: _____

2 Simplify:
(1 3/4) / (7/8)

Answer: _____

3 Simplify:
(4 1/2) / 3

Answer: _____

4 Simplify:
(5/6) / (2 1/2)

Answer: _____

5 Ty simplified (1 2/3) / (5/6) like this:
1 2/3 = 5/3, and 5/3 × 5/6 = 25/18.
What went wrong?

- ◯ A. Nothing. 25/18 is right.
- ◯ B. He should flip the TOP fraction: 3/5 × 5/6 = 1/2.
- ◯ C. 1 2/3 should be 3/3, not 5/3.
- ◯ D. He didn't flip the bottom fraction. It's 5/3 × 6/5 = 2.

6 Write 3 3/4 as an improper fraction.

Answer: _____

7 Write 2 2/5 as an improper fraction.

Answer: _____

Owen's Extra Practice (continued)

8 3/4 ÷ 3/8 = ?

Answer: _____

9 2 1/4 ÷ 1 1/2 = ?

Answer: _____

10 A trail-mix recipe uses 3/4 cup of oats per batch.
How many batches can you make with 4 1/2 cups of oats?

Answer: _____

11 x: 2, 5, 8
y: 14, 35, 56
The table is proportional (y = kx). What is k?

Answer: _____

12 A printer prints 135 pages in 9 minutes.
What is the unit rate in pages per minute?

Answer: _____

13 Each camper gets 3 marshmallows. y = marshmallows, x = campers, so y = 3x.
What is the constant of proportionality?

- ◯ A. k = 1/3
- ◯ B. k = 3
- ◯ C. k = x
- ◯ D. k = y

14 A proportional graph goes through the point (4, 10).
What is k? (Fraction, mixed number, or decimal is OK.)

Answer: _____

15 2 1/2 pounds of apples cost $5.50.
What is the price per pound?

Answer: $ _____

Owen's Extra Practice (continued)

16 Maya walks 3/4 mile in 1/4 hour.
What is her speed in miles per hour?

Answer: _____

Answer Key (grown-ups)

1. **3 1/3** — Complex fractions
Top: 2 1/2 = 5/2. Divide by the bottom = multiply by its reciprocal: 5/2 × 4/3 = 20/6 = 10/3 = 3 1/3.

2. **2** — Complex fractions
1 3/4 = 7/4. 7/4 × 8/7 = 56/28 = 2.

3. **1 1/2** — Complex fractions
4 1/2 = 9/2. Dividing by 3 = multiplying by 1/3: 9/2 × 1/3 = 9/6 = 3/2 = 1 1/2.

4. **1/3** — Complex fractions
2 1/2 = 5/2. 5/6 × 2/5 = 10/30 = 1/3.

5. **D. He didn't flip the bottom fraction. It's 5/3 × 6/5 = 2.** — Complex fractions
To divide by 5/6, multiply by its reciprocal 6/5: 5/3 × 6/5 = 30/15 = 2. His mixed number (5/3) was right.

6. **15/4** — Mixed numbers & dividing fractions
Whole × denominator + numerator: 3 × 4 + 3 = 15. Keep the denominator: 15/4.

7. **12/5** — Mixed numbers & dividing fractions
2 × 5 + 2 = 12, so 12/5.

8. **2** — Mixed numbers & dividing fractions
Keep, change, flip: 3/4 × 8/3 = 24/12 = 2.

9. **1 1/2** — Mixed numbers & dividing fractions
Make improper fractions first: 9/4 ÷ 3/2 = 9/4 × 2/3 = 18/12 = 3/2 = 1 1/2.

10. **6** — Mixed numbers & dividing fractions
Batches = 4 1/2 ÷ 3/4 = 9/2 × 4/3 = 36/6 = 6.

11. **7** — Unit rate & constant of proportionality (y = kx)
k = y ÷ x for any pair: 14 ÷ 2 = 7. Check: 56 ÷ 8 = 7.

12. **15** — Unit rate & constant of proportionality (y = kx)
Unit rate = pages ÷ minutes = 135 ÷ 9 = 15 pages per minute.

13. **B. k = 3** — Unit rate & constant of proportionality (y = kx)
k is the number that multiplies x (the amount for ONE camper): y = 3x, so k = 3. x is a variable, not the constant.

14. **2 1/2** — Unit rate & constant of proportionality (y = kx)
k = y ÷ x = 10 ÷ 4 = 5/2 = 2 1/2 (or 2.5).

15. **$2.20** — Unit rate & constant of proportionality (y = kx)
Price per pound = 5.50 ÷ 2.5 = $2.20.

16. **3** — Unit rate & constant of proportionality (y = kx)
Unit rate = (3/4) / (1/4), a complex fraction: 3/4 × 4/1 = 3 miles per hour.